

53. (a) We know that, the angles in the same segment of a circle are equal.

$$\therefore \angle XBY = \angle XAY = 45^\circ$$

In $\triangle BXY$, by angle sum property of a triangle

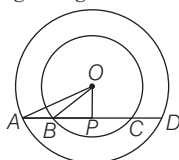
$$\angle BXY + \angle XBY + \angle BYX = 180^\circ$$

$$\Rightarrow 50^\circ + 45^\circ + \angle BYX = 180^\circ$$

$$[\because \angle BXY = 50^\circ]$$

$$\Rightarrow \angle BYX = 180^\circ - 95^\circ = 85^\circ$$

54. (a) In right angled $\triangle OPB$,



[by pythagoras theorem]

$$OB^2 = OP^2 + BP^2$$

$$\Rightarrow (15)^2 = (12)^2 + BP^2$$

$$\Rightarrow 225 - 144 = BP^2$$

$$\Rightarrow (BP)^2 = 81 \Rightarrow BP = 9 \text{ cm}$$

and in right angled $\triangle AOP$, by pythagoras theorem,

$$(OA)^2 = (OP)^2 + (AP)^2$$

$$\Rightarrow (20)^2 = (12)^2 + (AP)^2$$

$$\Rightarrow (AP)^2 = 400 - 144 = 256$$

$$\Rightarrow AP = 16 \text{ cm}$$

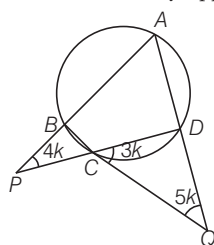
$$\text{Hence, } AB = AP - BP = 16 - 9 = 7 \text{ cm}$$

55. (b) Given, $\frac{x}{3} = \frac{y}{4} = \frac{z}{5} = k$ [say]

$$\therefore x = 3k, y = 4k \text{ and } z = 5k$$

$$\text{Since, } \angle DCQ = \angle BCP = 3k$$

[vertically opposite angle]



In $\triangle DCQ$, by angle sum property of a triangle

$$\angle CDQ = 180^\circ - (3k + 5k) = 180^\circ - 8k$$

By properties of cyclic quadrilateral,

$$\angle CDQ = \angle CBA = 180^\circ - 8k$$

$$\Rightarrow \angle PBC = 8k$$

In $\triangle PBC$, by angle sum property of a triangle

$$\angle P + \angle B + \angle C = 180^\circ$$

$$\therefore 4k + 8k + 3k = 180^\circ$$

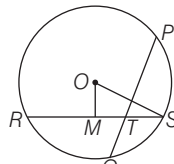
$$\Rightarrow k = \frac{180^\circ}{15} \Rightarrow k = 12^\circ$$

$$\therefore x = 36^\circ, y = 48^\circ, z = 60^\circ$$

56. (b) We know that, angle subtended on the circumference is half of the angle subtended at the centre.

$$\angle AQP = \frac{1}{2} \angle AOP = \frac{1}{2} 75^\circ = 37.5^\circ$$

57. (b)



Given, $OM = 3 \text{ cm}$, $OS = 5 \text{ cm}$.

$$MS = 4 \text{ cm} = RM$$

[Using pythagoras theorem]

$$\Rightarrow RS = 8 \text{ cm}$$

We know that, $TS = 1/3$ of RT

$$\Rightarrow TS = 1/4 \text{ of } RS, \text{ if } RS = 8 \text{ cm}$$

Then, $TS = 2 \text{ cm}$

$$\text{Also, } RT \times TS = PT \times TQ$$

[When there are two intersecting chords, the product of the segments of one chord is equal to the product of the segments of other.]

$$\Rightarrow 6 \times 2 = PT \times TQ$$

$$\Rightarrow PT \times TQ = 12$$

Now, by $AM \geq GM$ inequality,

$$(PT + TQ) / 2 \geq \sqrt{(PT \times TQ)}$$

$$(PT + TQ) / 2 \geq \sqrt{12}$$

$$\Rightarrow PT + TQ \geq 2\sqrt{12} \Rightarrow PQ \geq 2\sqrt{12}$$

$$\text{or } PQ \geq 4\sqrt{3}$$

$$\therefore \text{Minimum } PQ = 4\sqrt{3}$$

58. (a) When 2 chords AB and CD intersect at P then

$$AP \times PB = CP \times PD$$

Hence,

$$4 \times 6 = 3 \times PD, \text{ Thus,}$$

$$PD = 8 \text{ cm}$$

$$\text{Now } AB = AP + PB$$

$$= 10 \text{ cm}$$

$$\text{and } CD = CP + PD$$

$$\text{Thus, } CD = 11 \text{ cm}$$

Consider the circle with center O .

Drop a perpendicular from O to chord AB and CD . This will bisect the chords at X and Y .

$$\therefore AX = \frac{AB}{2} = 5 \text{ cm}$$

$$\text{and } YD = \frac{CD}{2} = \frac{11}{2} \text{ cm}$$

$$\text{Now, } PX = AX - AP = (5 - 4) = 1 \text{ cm}$$

$$OY = PX = 1 \text{ cm}$$

[$\because PXOY$ is a rectangle]

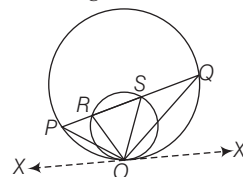
In right triangle OYD

$$OD = \sqrt{OY^2 + YD^2}$$

[by pythagoras theorem]

$$= \sqrt{1^2 + \left(\frac{11}{2}\right)^2} = \frac{5\sqrt{5}}{2} \text{ cm}$$

59. (a) In the figure,



Let $\angle POX = x^\circ$, $\angle ROP = y^\circ$

$$\angle POX = \angle PQO = x$$

$$\text{and } \angle ROX = \angle RSO = x^\circ + y^\circ$$

[angles in alternate segment]

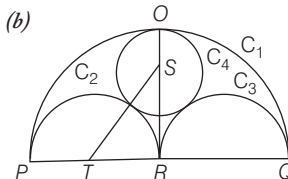
Now, in $\triangle SQO$, $\angle RSO$ is the exterior angle

$$\therefore \angle SQO + \angle SOQ = x^\circ + y^\circ$$

$$\Rightarrow x + \angle SOQ = x^\circ + y^\circ$$

$$\Rightarrow \angle SOQ = y = \angle ROP$$

60. (b)



Assume that the radius of $C_4 = r$ and $PQ = k$

$$\text{Now, } PR = k/2 = RQ = RO$$

$$\Rightarrow RS = (k/2) - r, RT = k/4$$

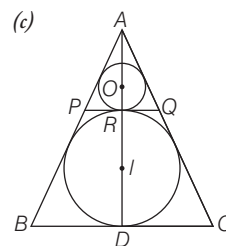
$$ST = (k/4) + r$$

Applying pythagoras theorem in triangle STR .

$$\left(\frac{k}{4} + r\right)^2 = \left(\frac{k}{4}\right)^2 + \left(\frac{k}{2} - r\right)^2$$

$$\Rightarrow r = k/6 = PQ/6$$

61. (c)



In-radius of equilateral triangle of side

$$a = \frac{a}{2\sqrt{3}}$$

$$\text{Diameter of larger circle} = \frac{a}{\sqrt{3}}$$

Let us say common tangent PQ touches the two circle at R , centre of smaller circle is I .

Now, PQ is parallel to BC . AR is perpendicular to PQ . Triangle PQR is also an equilateral triangle and $AORID$ is a straight line.

$$AD = \frac{\sqrt{3}}{2}a, RD = \frac{a}{\sqrt{3}}, AR = \frac{\sqrt{3}}{2}a - \frac{a}{\sqrt{3}}$$

$$AR = \frac{3a - 2a}{2\sqrt{3}} = \frac{a}{2\sqrt{3}}, AR = \frac{1}{3}AD$$

Radius of smaller circle = $1/3$ radius of larger circle

Radius of smaller circle

$$= \frac{1}{3} \cdot \frac{a}{2\sqrt{3}} = \frac{a}{6\sqrt{3}}$$

Area of smaller circle = πr^2

$$\pi \left(\frac{a}{6\sqrt{3}} \right)^2 = \frac{\pi a^2}{108}$$

$$\text{Area of triangle} = \frac{\sqrt{3}}{4} a^2$$

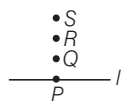
$$\text{Ratio} = \frac{\pi a^2}{108} : \frac{\sqrt{3}}{4} a^2, \pi : 27\sqrt{3}$$

62. (d) I. It is true that the opposite angles of a cyclic quadrilateral are supplementary.

II. It is also true that the angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

Hence, both statements are individually true, but neither statements implies to each other.

63. (c) I. It is clear from the figure that points Q, S and R are in a straight line.

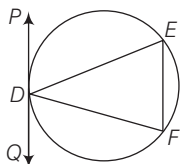


So, statement I is true.

II. Since, PQ is parallel to EF .

$$\therefore \angle PDE = \angle DEF \text{ [alternate angle]}$$

$$\text{Also, } \angle PDE = \angle EFD$$



[angle in the alternate segment of chord ED]

$$\therefore \angle DEF = \angle DFE$$

$$\Rightarrow \triangle DEF \text{ is an isosceles triangle.}$$

So, statement II is also true.

Hence, the statement I and II are correct.

64. (d) As, the tangents drawn from an external point to a circle are equal.

$$\therefore AP = AR \dots (i)$$

$$BQ = BP \dots (ii)$$

$$\text{and } CR = QC \dots (iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$AP + BQ + CR = BP + QC + RA$$

and perimeter of

$$\triangle ABC = AB + BC + CA$$

$$= (AP + PB) + (BQ + QC) + (CR + RA)$$

$$= (AP + BQ) + (BQ + CR) + (CR + AP)$$

$$= 2(AP + BQ + CR)$$

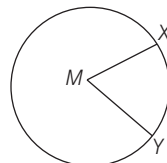
$$\therefore AP + BQ + CR = 1/2$$

[perimeter of $\triangle ABC$]

Hence, the statement I and II are correct.

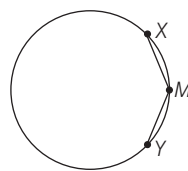
65. (d) I. Consider the possibility that M

could be the centre of the circle. We know this could be true because M being the center point of the circle would make line XM and YM radii of the circle, which would mean that they are equal.



II. The statement presents us with the possibility that

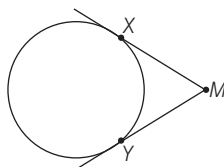
point M lies somewhere on the arc of XY . Well, if point M rested exactly halfway



between X and

Y , then straight lines drawn from X to M and Y to M would certainly be equal.

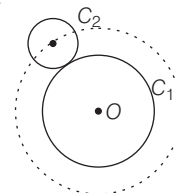
III. The statement presents us with the possibility that M lies somewhere outside of the circle. If two tangents are drawn from point M at X and Y they will be equal.



Hence, the statement I, II and III are correct.

66. (d) I. Circle is a locus of points which are equidistant from a fixed point.

II. Let C_1 be a fixed circle and C_2 be moving circle then locus of centre of C_2 is also a circle.



III. There does not exist any point which is equidistant from three distinct points on a line.

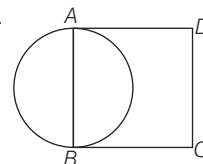
IV. If P lies on bisector of $\angle B$ then P is equidistant from AB and BC also, P lies on bisector of $\angle C$.

$$\Rightarrow P \text{ is equidistant from } BC \text{ and } CD$$

$$\Rightarrow P \text{ is equidistant from } AB \text{ and } CD.$$

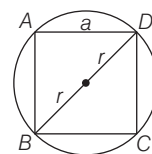
Hence, I, II, and IV are correct statement.

67. (d) I.



So, the square is not cyclic.

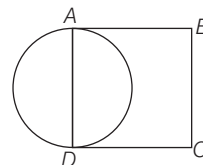
II.



Square has less area than circle.

III. From the above figure it is clear that the diameter of circle is equal to the diagonal of square. $\Rightarrow 2r = a\sqrt{2}$

IV.



So, it is possible that the sides of square is equal to diameter of circle.

Hence statement II, III and IV are correct.

68. (c) We observe that the arc BC makes $\angle BOC = z$ at the centre and $\angle BAC = x$ at a point on the circumference.

$$\therefore z = 2x$$

In $\triangle OBC$, we have

$$\angle OBC + \angle OCB + \angle BOC = 180^\circ$$

$$\Rightarrow y + z + y = 180^\circ$$

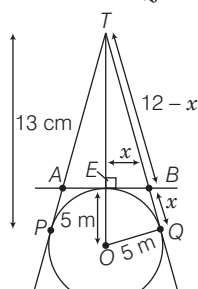
$$\Rightarrow z + 2y = 180^\circ \Rightarrow 2x + 2y = 180^\circ$$

$$\Rightarrow x + y = 90^\circ$$

$$\Rightarrow \angle BAC + \angle OBC = 90^\circ$$

69. (a) In $\triangle OBC$, we have
 $\angle OBC + \angle OCB + \angle BOC = 180^\circ$
 $\Rightarrow y + y + t = 180^\circ \Rightarrow t = 180^\circ - 2y$
 Now, $z = 360^\circ - t \Rightarrow z = 360^\circ - (180^\circ - 2y)$
 $\Rightarrow 2x = 180^\circ + 2y \Rightarrow 2x - 2y = 180^\circ$
 $[\because z = 2x]$
 $\Rightarrow x - y = 90^\circ$
 $\Rightarrow \angle BAC - \angle OBC = 90^\circ$

70. (b) In right angled $\triangle OTQ$,
 $OT^2 = OQ^2 + TQ^2$



$$\Rightarrow (13)^2 = (5)^2 + (TQ)^2$$

$$\Rightarrow TQ^2 = 169 - 25 = 144 \Rightarrow TQ = 12 \text{ cm}$$

In right angled $\triangle TEB$,

$$TB^2 = EB^2 + TE^2$$

$$\because (EB = BQ)$$

$$\Rightarrow (12 - x)^2 = BQ^2 + TE^2$$

$$\Rightarrow 144 + x^2 - 24x = x^2 + (8)^2$$

$$\Rightarrow 144 + x^2 - 24x = x^2 + 64$$

$$\Rightarrow 24x = 80 \Rightarrow x = \frac{20}{6} = \frac{10}{3} \text{ cm}$$

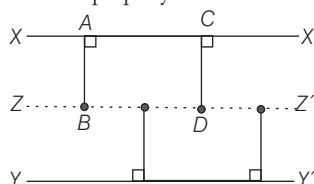
$$\therefore AB = 2EB = 2x = 2 \times \frac{10}{3} = \frac{20}{3} \text{ cm}$$

Hence, the length of AB is $\frac{20}{3}$ cm.

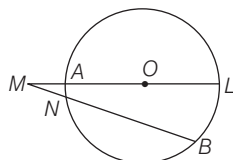
71. (b) The locus obtained is the circumference of the circle concentric with the given circle and having radius equal to the distance of the chords from the centre.

72. (c) Statements I and II are both correct, because the locus of points which are equidistant from two parallel lines is a line parallel to both of them and lie mid way between them.

Also, it is true that the perpendicular distances of any point on this locus line from two original parallel lines are equal. further, no point outside this locus line has this property.



73. (d) Since, secants LA and BN are intersecting at an exterior point M , then



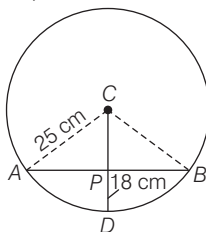
$$LM \times AM = BM \times NM$$

$$\Rightarrow \frac{MA}{MB} = \frac{MN}{LM} < 1$$

74. (b) The perpendicular bisector of the chord of a circle always passes through the centre. So, statement I is wrong.

The angle in a semi-circle is a right angle. So, statement II is correct.

75. (d) Here, $CA = CD = CB = 25$ cm



and $DP = 18$ cm

In right angled $\triangle ACP$,

$$CP = CD - PD = 25 - 18 = 7 \text{ cm}$$

Now, $AC^2 = CP^2 + AP^2$

[by pythagoras theorem]

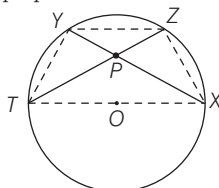
$$\therefore AP = \sqrt{AC^2 - CP^2} = \sqrt{(25)^2 - (7)^2}$$

$$= \sqrt{625 - 49} = \sqrt{576} = 24 \text{ cm}$$

Similarly, $PB = 24$ cm

$$\therefore AB = AP + PB = 24 + 24 = 48 \text{ cm}$$

76. (c) When two chords of a circle intersect internally, then they are divided in a proportion.



$$\text{i.e. } PX \cdot PY = PZ \cdot PT$$

In $\triangle PXZ$ and $\triangle PTY$, $\angle ZPX = \angle YPT$

[vertically opposite angles]

$$\angle PZX = \angle PYT$$

[angles in same segment]

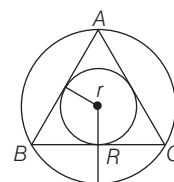
$$\angle PXZ = \angle PTY$$

[angles in same segment]

$$\therefore \triangle PXZ \sim \triangle PTY$$

Hence, the both statements are correct.

77. (a) In $\triangle ABC$, $AB = BC = AC$



Hence, $\triangle ABC$ is equilateral triangle.

Let r be the radius of incircle and R be the radius of circumcircle.

$$\text{Now, radius of incircle, } r = \frac{\text{side}}{2\sqrt{3}} = \frac{AB}{2\sqrt{3}}$$

and radius of circumcircle,

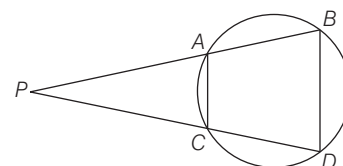
$$R = \frac{\text{side}}{\sqrt{3}} = \frac{AB}{\sqrt{3}}$$

$$\text{So, the required ratio} = \frac{R}{r} = \frac{AB / \sqrt{3}}{AB / 2\sqrt{3}}$$

$$= \frac{2}{1} = 2 : 1$$

78. (b) I. AB and CD are chords when produced meet externally at P .

$$\therefore AP \times BP = CP \times DP$$



II. Since, $ABDC$ is a cyclic quadrilateral

$$\angle PAC = \angle PDB$$

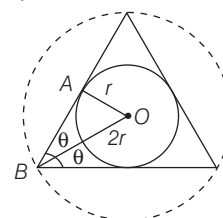
$$\text{and } \angle PCA = \angle PBD$$

Also, $\angle APC = \angle BPD$ [common]

$$\therefore \triangle PAC \sim \triangle PDB$$

So, statement II is true.

79. (c) Let $OA = r$ be the inradius of circle. Then, circumradius = $OB = 2r$



We know that, inradius is the perpendicular distance of centre O from side of triangle and circumradius OB bisect $\angle B$. Again, let $\angle OBA = \theta$

In right angled $\triangle OAB$,

$$\sin \theta = \frac{OA}{OB} = \frac{r}{2r}$$

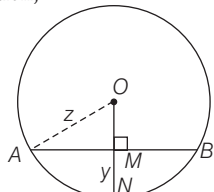
$$\Rightarrow \sin \theta = \frac{1}{2} = \sin 30^\circ \quad \left[\because \sin 30^\circ = \frac{1}{2} \right]$$

$$\therefore \theta = 30^\circ$$

$$\text{Then, } \angle B = 2 \times \theta = 2 \times 30^\circ = 60^\circ$$

Hence, given triangle is an equilateral triangle.

80. (d) Let O be the centre of circle and AB be the chord of an arc. According to the question,



Length of chord $AB = x$, Radius of circle $= OA = z$
and height of arc $MN = y$
Now, $AM = MB = \frac{AB}{2} = \frac{x}{2}$

[Perpendicular from the centre to a chord bisects the chord]
and $OM = ON - MN$
 $= z - y$ [$\because ON = z$ (radius)]

In right angled $\triangle OMA$,
 $OA^2 = OM^2 + AM^2$
[by pythagoras theorem]

$$\Rightarrow z^2 = (z - y)^2 + \left(\frac{x}{2}\right)^2$$

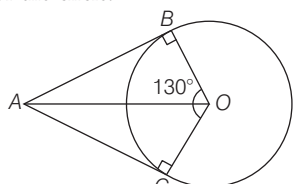
$$\Rightarrow z^2 = z^2 + y^2 - 2yz + \frac{x^2}{4}$$

$$[\because (a - b)^2 = a^2 + b^2 - 2ab]$$

$$\Rightarrow 2yz - y^2 = \frac{x^2}{4} \Rightarrow 4(2yz - y^2) = x^2$$

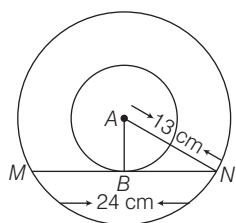
$$\Rightarrow 4y(2z - y) = x^2$$

81. (c) Given, AB and AC are the tangents of the circle.



and $OB = OC$ [radius of circle]
 $\therefore \angle B = \angle C = 90^\circ$
[radius is perpendicular to the tangent]
In quadrilateral $ABOC$,
 $\angle ABO + \angle OCA + \angle BAC + \angle BOC = 360^\circ$
 $\Rightarrow 90^\circ + 90^\circ + \angle BAC + 130^\circ = 360^\circ$
 $\Rightarrow \angle BAC = 180^\circ - 130^\circ = 50^\circ$
Hence, the angle between the tangents at the ends of the radius is 50° .

82. (a) Given, $MN = 24$ cm and
 $AN = \frac{26}{2}$ cm $= 13$ cm



Now, we draw a perpendicular bisector from A to MN , which meets MN at B .

Then, $MB = BN = \frac{MN}{2} = \frac{24}{2} = 12$ cm

[$\because$ perpendicular from centre to the chord bisects the chord]

In right angled $\triangle ABN$,

$$AN^2 = AB^2 + BN^2$$

$$\Rightarrow 13^2 = AB^2 + 12^2$$

$$\Rightarrow 169 = AB^2 + 144$$

$$\Rightarrow AB^2 = 169 - 144 = 25$$

$$\Rightarrow AB = 5$$
 cm

$\therefore$ Radius of the inner circle is 5 cm.

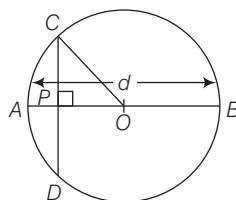
83. (d) Given, $AD = 34$ cm and $AB = 30$ cm
Let C be a point on AB such that $OC \perp AB$

$$\Rightarrow OD = OB = 17$$
 cm
Also, $CB = \frac{AB}{2} = 15$ cm
[Perpendicular from centre to the chord bisects the chord]
$$\Rightarrow OC = \sqrt{OB^2 - CB^2} = \sqrt{17^2 - 15^2}$$

$$= \sqrt{289 - 225} = \sqrt{64} = 8$$
 cm

84. (d) Since, $AP = BP$; $DE = BD$ and $AC = CE$
 $\therefore$ Perimeter of $\triangle PCD$
 $= PC + PD + CD$
 $= PC + PD + CE + DE$
 $= (PC + CE) + (PD + DE)$
 $= (PC + CA) + (PD + BD)$
 $= PA + PB = 2PA = 2 \times 16 = 32$ cm

85. (c) Given, $AB = d$ is diameter of circle and CD intersect AB at P in the ratio 1 : 2.



$$\therefore AP = \frac{d}{3} \text{ and } PB = \frac{2d}{3}$$

$$\Rightarrow OA = OC = OB = d/2$$

$$\left[\because \text{radius of circle} = \frac{1}{2} \text{ diameter} \right]$$

In $\triangle OPC$, $OP = OA - AP = \frac{d}{2} - \frac{d}{3} = \frac{d}{6}$

Applying pythagoras theorem in $\triangle OPC$,

$$OC^2 = OP^2 + PC^2$$

$$\Rightarrow \left(\frac{d}{2}\right)^2 = \left(\frac{d}{6}\right)^2 + PC^2$$

$$\Rightarrow (PC)^2 = \frac{d^2}{4} - \frac{d^2}{36}$$

$$= \frac{9d^2 - d^2}{36} = \frac{8d^2}{36}$$

$$\Rightarrow PC = \frac{2\sqrt{2}d}{6} = \frac{\sqrt{2}d}{3}$$

$$\therefore CD = 2PC = \frac{2\sqrt{2}d}{3}$$

86. (b) 1 km = 1000 m, 1 dm = 10 m
 $\therefore$ 2 km and 26 dm
 $= 2000 + 260 = 2260$ m

Let radius be r m.

Distance covered in revolution $= 2\pi r$

Distance covered in 452 revolution

$$= 452 \times 2\pi r$$

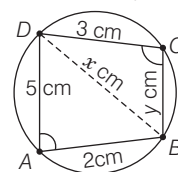
$$\therefore 452 \times 2\pi r = 2260$$

$$\therefore r = \frac{2260 \times 7}{452 \times 2 \times 22} = \frac{35}{44}$$

$$\text{Diameter} = 2 \times \frac{35}{44} = \frac{35}{22} = 1\frac{13}{22}$$
 m

87. (b) Distance covered in 1 revolution
 $= 2\pi (0.3)$ m $= 0.6\pi$ m
 $\therefore$ Distance covered in 140 revolutions
 $= 140 \times 0.6\pi$ m
 $\Rightarrow$ Speed the cycle
 $= 140 \times 0.6 \times \frac{22}{7} \times 60$ m / h
 $= 15840$ m/h $= 15.84$ km/h

88. (a) Let $ABCD$ be cyclic quadrilateral.



in which $\angle A = 60^\circ$,

$$AB = 2 \text{ cm}, AD = 5 \text{ cm}$$

Since, $\angle A + \angle C = 180^\circ$

[property of cyclic quadrilateral]

$$\Rightarrow 60^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 180^\circ - 60^\circ = 120^\circ$$

In $\triangle ABD$, let BD be x cm.

$$\cos 60^\circ = \frac{2^2 + 5^2 - x^2}{2 \times 2 \times 5}$$

[by cosine formula]

$$\Rightarrow \frac{1}{2} = \frac{4 + 25 - x^2}{2 \times 2 \times 5} \Rightarrow 29 - x^2 = 10$$

$$\Rightarrow x^2 = 29 - 10 \Rightarrow x^2 = 19$$

Again, in $\triangle BCD$,

$$\cos 120^\circ = \frac{3^2 + y^2 - x^2}{2 \times 3 \times y}$$

[by cosine formula]

Let length of fourth side of cyclic quadrilateral be y cm.

$$\Rightarrow -\frac{1}{2} = \frac{9 + y^2 - 19}{2 \times 3 \times y} \quad [\because x^2 = 19]$$

$$\Rightarrow -3y = 9 + y^2 - 19$$

$$\begin{aligned}
 &\Rightarrow y^2 + 3y - 10 = 0 \\
 &\Rightarrow y^2 + 5y - 2y - 10 = 0 \\
 &\Rightarrow y(y + 5) - 2(y + 5) = 0 \\
 &\Rightarrow (y + 5)(y - 2) = 0 \\
 &\Rightarrow y + 5 = 0 \text{ or } y - 2 = 0 \\
 &\therefore y = 2 \text{ cm}
 \end{aligned}$$

89. (c) Since, two circles touch each other externally if distance between their centres = Sum of their radii

$$O_1O_2 = r + \frac{a}{2} \text{ and } O_2M = a - r$$

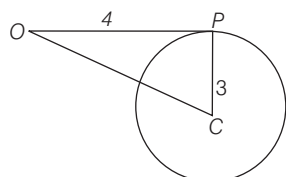
$$O_1M = \frac{a}{2}$$

In right $\triangle O_1MO_2$

$$\begin{aligned}
 (O_1O_2)^2 &= (O_1M)^2 + (O_2M)^2 \\
 \Rightarrow \left(r + \frac{a}{2}\right)^2 &= \frac{a^2}{4} + (a - r)^2
 \end{aligned}$$

$$\begin{aligned}
 &\Rightarrow 3r = a \\
 &\therefore r = \frac{a}{3}
 \end{aligned}$$

90. (c) Since, OP is tangent to the circle $OP \perp PC$.



Here, $\triangle OPC$ is right angled triangle with $\angle P = 90^\circ$

$$OC = \sqrt{(3)^2 + (4)^2} = 5 \text{ units}$$

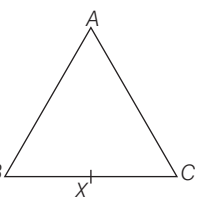
$$\therefore \sin(\angle COP) = \frac{3}{5}$$

91. (d) In $\triangle ABC$, $AB = AC$ i.e. $\triangle ABC$ is isosceles triangle.

If L is the locus of points X inside or on the triangle, such that $BX = CX$. Then, L is a straight line passing through A .

As, AX is bisecting the line BC , then it passes through the centroid.

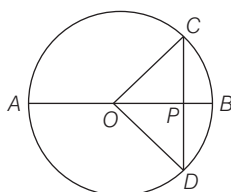
As, $AB = AC$ in $\triangle ABC$, then AX is perpendicular to BC , hence it passes through the orthocentre.



In $\triangle ABC$, $AB = AC$ and $AX \perp BC$. So, AX is angle bisector of $\angle A$, hence it passes through the incentre. Hence, all statements are correct.

92. (b) Given, radius of a circle = 2 unit

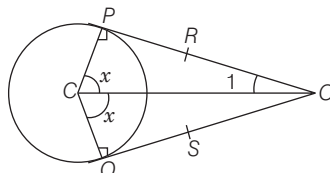
$$\therefore CP = PD = \frac{CD}{2} = \frac{2}{2} = 1 \text{ unit}$$



$$\begin{aligned}
 \text{Now, } OP &= OD^2 - PD^2 \\
 &= \sqrt{4 - 1} = \sqrt{3} \text{ units}
 \end{aligned}$$

$$\therefore AP = AO + OP = (2 + \sqrt{3}) \text{ units}$$

93. (c) In the given figure, OP and OQ are tangents to the circle with centre C .



Then, $OP = OQ$ and

$$\angle CPO = \angle CQO = 90^\circ$$

$$\begin{aligned}
 \text{I. We have, } OR \times SQ &= OS \times RP \\
 &\Rightarrow (OP - RP) \times SQ \\
 &= (OQ - SQ) \times PR \\
 &\Rightarrow (OP - RP) \times SQ \\
 &= (OP - SQ) \times PR \\
 &\Rightarrow SQ = PR \Rightarrow OR = OS
 \end{aligned}$$

Hence, statement I is correct.

$$\text{II. } \therefore \triangle OPC \cong \triangle OCQ$$

$$\text{In } \triangle OPC, \quad 90^\circ + x + \angle 1 = 180^\circ$$

$$\Rightarrow x + \angle 1 = 90^\circ$$

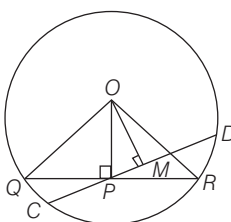
$$\Rightarrow \angle POC + \angle QCO = x + \angle 1 = 90^\circ$$

Hence, statement II is correct.

94. (a) We have, $MP = \sqrt{24}$ cm,

$$OQ = OR = 7 \text{ cm}$$

$$\text{and } QP = PR = \frac{QR}{2} = \frac{2}{2} = 1$$



$$\begin{aligned}
 \text{I. In } \triangle OPQ, \quad OP &= \sqrt{OQ^2 - QP^2} \\
 &= \sqrt{49 - 1} = \sqrt{48} = 4\sqrt{3} \text{ cm}
 \end{aligned}$$

$$\text{In } \triangle OMP, OP^2 = PM^2 + OM^2$$

$$\Rightarrow OM = \sqrt{48 - 24} = \sqrt{24} \text{ cm}$$

$$\text{So, } OM = PM = \sqrt{24} \text{ cm}$$

$$\therefore \angle OPM = \angle POM = 45^\circ$$

$$\begin{aligned}
 \text{Now, } \angle QPD &= 90^\circ + \angle OPM \\
 &= 90^\circ + 45^\circ = 135^\circ
 \end{aligned}$$

Hence, statement I is correct.

- II. Now, if $CP = m$ and $PD = n$, then $QP \times PR = CP \times PD$

$$\Rightarrow 1 \times 1 = m \times n \Rightarrow mn = 1$$

Also, $CM = MD$

$$\Rightarrow CP + PM = PD - PM$$

$$\Rightarrow m + \sqrt{24} = n - \sqrt{24}$$

$$\Rightarrow n - m = 2\sqrt{24}$$

$$\begin{aligned}
 \therefore m + n &= \sqrt{(m - n)^2 + 4mn} \\
 &= \sqrt{96 + 4} = 10
 \end{aligned}$$

So, statement II is also true

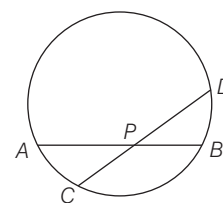
$$\begin{aligned}
 \text{III. } \frac{\text{ar}(\triangle OPR)}{\text{ar}(\triangle OMP)} &= \frac{\frac{1}{2} \times OP \times PR}{\frac{1}{2} \times OM \times PM} \\
 &= \frac{4\sqrt{3} \times 1}{\sqrt{24} \times \sqrt{24}} = \frac{1}{2\sqrt{3}}
 \end{aligned}$$

So, statement III is not true.

Hence, the statement I, II are correct.

95. (c) From the given circle,

$$AP \times BP = CP \times DP \quad \dots(i)$$



Given that, $A'P' = AP$,

$B'P' = BP$, $C'Q' = CP$ and $D'Q' = DP$

Now,

$$\begin{aligned}
 \text{ar}(\triangle A'P'B') &= \frac{1}{2} \times A'P' \times P'B' \\
 \text{ar}(\triangle C'Q'D') &= \frac{1}{2} \times C'Q' \times Q'D'
 \end{aligned}$$

$$= \frac{AP \times BP}{CP \times DP} = \frac{AP \times BP}{AP \times BP} = 1 \quad [\text{using (i)}]$$

$\therefore \triangle A'P'B'$ and $\triangle C'Q'D'$ are triangles of same area.

So, statement III cannot be false.

Hence, only (c) is correct option.